

A tunable weak-field applicator for testing radical-pair ROS modulation in exercise recovery

Build specification and self-controlled experimental protocol

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July 4, 2026

Read first — status and safety. This is a *research instrument*, not a medical device or a treatment. It is built to *measure* whether a weak magnetic field changes recovery markers, not to deliver a promised benefit. Static fields up to 3 mT are $\sim 60\times$ Earth’s field and far below a fridge magnet or MRI, with no established acute hazard; nonetheless, do not use with any implanted electronic device (pacemaker, pump), and any protocol on other people requires ethics oversight. The predicted effect is a *few percent* and unproven in living humans. Crucially, suppressing exercise ROS is **not** automatically beneficial (Sec. 8).

1 Purpose

The companion note [1] models the mitochondrial flavin–superoxide radical pair as a spin gate whose triplet (superoxide-escape) yield depends on a weak applied magnetic field, on the scale of the flavin hyperfine coupling ($\sim 0.2\text{--}3$ mT). Usselman et al. [2] measured exactly such a millitesla-scale shift in the cellular $\text{O}_2^{\bullet-}/\text{H}_2\text{O}_2$ ratio. This document specifies a low-cost apparatus that produces a *tunable, uniform* field over a limb, and a *sham-controlled* protocol to test whether scanning that field measurably affects post-exercise recovery.

2 Field applicator (Helmholtz pair)

A Helmholtz pair gives a uniform field over a central region large enough for a forearm or calf. With coil radius R and coil separation equal to R , the central field is

$$B = \left(\frac{4}{5}\right)^{3/2} \frac{\mu_0 N I}{R}. \quad (1)$$

For $R = 0.15$ m (30 cm coils, 15 cm apart) the uniform region is ~ 9 cm across. Table 1 gives the current for each target field at $N = 100$ turns per coil.

target B	ampere-turns NI	current I ($N=100$)
0.2 mT (near the effect peak)	33	0.33 A
1.0 mT	167	1.67 A
3.0 mT (upper scan)	500	5.00 A

Table 1: Helmholtz drive currents, $R = 0.15$ m.

Concrete build. Two coils of $N = 100$ turns of AWG20 (0.5 mm²) enamelled copper on 30 cm formers. Wire length ≈ 188 m, coil-pair resistance $\approx 6.3\ \Omega$. At the 3 mT maximum the pair dissipates ≈ 160 W; a typical protocol runs at ≤ 1 mT (≈ 20 W) from a 12 V supply. **Drive:**

a constant-current source (a buck/linear current regulator; the UC3843-class controller from common maker kits works) with a front-panel set-point for B . Keep the duty modest and let the coils warm-limit; add an NTC cutout at 45 °C.

3 Calibration

Do not trust the set-point dial—*measure* the field. A Hall-effect gaussmeter or a calibrated linear Hall sensor (e.g. a ± 10 mT ratiometric sensor) at the limb position maps current to B before any session. This step is what separated the honest instruments from the folklore devices in the wider “energy” literature: the number on the coil must be verified, not assumed.

4 Dose plan

Figure 1 shows the model’s predicted change in superoxide versus field, with recommended set-points to scan: 0 (sham/off), 0.2, 0.5, 1.0 and 3.0 mT. The predicted signal is largest near the 0.2–0.5 mT resonance and declines toward 3 mT—so the experiment should concentrate points at low field, not high.

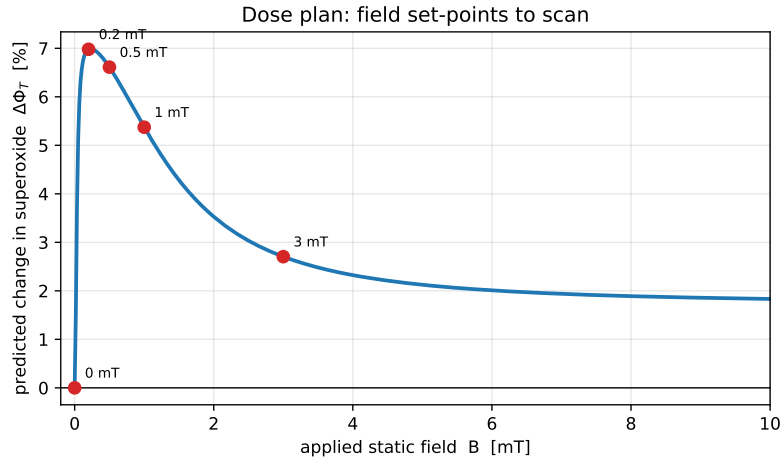


Figure 1: Predicted change in superoxide yield $\Delta\Phi_T$ vs applied field (radical-pair model, CISS $P = 0.35$); red points are the recommended experimental set-points. The effect peaks near 0.2–0.5 mT.

5 Sham-controlled self-experiment

The single most important design element is a **sham coil** that looks, sounds and warms identically but produces no net field (bifilar/counter-wound so the currents cancel). Sessions are **blinded** (a helper sets real/sham from a coded list; you do not know which) and **crossed over** (each condition repeated on matched days).

Recovery read-outs (choose ≥ 2 ; all low-cost, non-invasive):

- **HRV** (heart-rate variability, rMSSD) each morning from a chest strap or ring—the cleanest daily recovery proxy;
- **perceived soreness** (DOMS) on a 0–10 scale at +24 and +48 h;
- **force recovery**: countermovement-jump height or grip dynamometer at +24/ +48 h vs pre-exercise;
- optional **creatin kinase** (finger-prick point-of-care) as a muscle-damage marker.

Session flow (one condition): standardized run → apply coil to the worked limb for 30–60 min at the assigned B (or sham) → log read-outs at +24 and +48 h. Repeat each condition ≥ 5 times before comparing; a real few-percent physiological effect needs $\sim$ dozens of paired sessions to clear day-to-day noise, which is the honest cost of measuring something this small.

Analysis: paired comparison (real vs sham) per read-out; treat B as the dose axis of Fig. 1. A genuine effect appears as a *dose-shaped* response peaking at low field, not a flat offset (which would flag a placebo or a thermal artifact from the coil warming the limb—control for that with the sham’s matched heat).

6 The bifilar sham coil

The sham must be electrically and thermally identical to the real coil but produce *no net field*. This is achieved by winding it **bifilar**: two enamelled wires laid side by side and wound together as one, then connected so that the current runs *out* along wire 1 and *back* along wire 2. Every turn then carries $+I$ and $-I$ a millimetre apart, so the magnetic fields cancel to $< 1\%$ of the real coil, while the wire length, resistance and ohmic heating are the same (Fig. 2). Use the *same* wire, turn count and former as the real coil; the only difference is the connection at the ends.

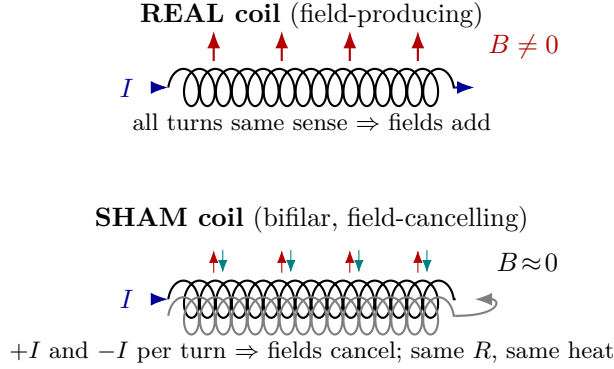


Figure 2: Real vs sham winding. Both use identical wire, turns and former; the sham is wound bifilar (two wires together, connected to counter-run the current) so its field cancels while its resistance and heating match the real coil—the control that separates a genuine field effect from placebo or coil-warming artifacts.

Winding recipe (per coil). (1) Cut two equal wire lengths (≈ 95 m each for $R = 15$ cm, $N = 100$). (2) Twist the two starts together loosely and wind them *as a pair*, N turns, onto the former—this is the bifilar layer. (3) At the finish, join the end of wire 1 to the end of wire 2 (the “U-turn”); the two free starts become the terminals. Current now flows out wire 1 through all turns and back wire 2 through the same turns: net field ≈ 0 , resistance = $2 \times$ one-wire (identical to the real coil’s series winding). Verify with the gaussmeter: the sham should read < 0.05 mT at the same current that gives 1 mT on the real coil.

7 Bill of materials

Table 2 is a realistic parts list (prices as of 2026, RU retail, order of magnitude; reuse of a bench supply and an HRV device you already own cuts the total substantially). A full new build is $\sim 12\,000$ – $18\,000$ RUB; the minimum functional build (reusing PSU + phone/watch HRV) is $\sim 5\,000$ RUB.

Item	Spec	Purpose	Price [RU]
Enamelled Cu wire	AWG20 (0.8 mm), ≥ 400 m	real + sham coils	2 000–3 000
Coil formers ($\times 2$)	30 cm dia. ring (PVC/plywood/3D-print)	winding support	500–1 200
Constant-current driver	buck CC-CV, ≥ 5 A (XL4015 / DPS-class)	set-point field control	500–900
DC supply	12 V, ≥ 10 A (or reuse)	power	0–3 000
Linear Hall sensor	± 10 mT ratiometric (or SS49E)	field calibration	300–700
Microcontroller	Arduino Nano / ESP32	current set-point, logging	400–900
NTC + comparator	45 °C cutout	thermal safety	200–400
Relay/DPDT switch	coded real/sham selection	blinding	300–600
Enclosure + connectors	—	assembly	800–1 500
Thermal pad/insulation	limb-side	comfort/safety	300–500
Core electronics + coils subtotal			5 600–13 100
HRV sensor	chest strap (Polar H10) or ring	recovery read-out	0–7 000
CK point-of-care (opt.)	finger-prick strips	muscle-damage marker	0–3 000
Dynamometer/jump mat (opt.)	grip or CMJ	force recovery	0–4 000

Table 2: Bill of materials. “0” entries are items commonly already owned (bench supply, a watch/ring with HRV). Gaussmeter calibration can also use a borrowed instrument.

8 Why “less ROS” is not automatically “better recovery”

Exercise-induced superoxide and H_2O_2 are *signalling* molecules: the acute burst drives mitochondrial biogenesis, endogenous antioxidant upregulation and the training adaptation itself (redox hormesis). Randomized trials show that blunting this burst with high-dose vitamin C/E can *impair* endurance adaptation and insulin sensitivity [3, 4]. Consequences for this instrument:

- a device that simply lowers ROS could *reduce* training gains;
- the plausible useful window is trimming a *pathological excess* (acute injury, overtraining, ischaemia-reperfusion) while sparing the adaptive signal—an open question the instrument is built to test;
- the right primary endpoint is therefore recovery *without* degraded adaptation over weeks, not acute ROS alone.

9 What a result would mean

A reproducible, dose-shaped, sham-beating shift in a recovery read-out would be the first human-level evidence that CISS-tuned radical-pair spin chemistry is a controllable lever on exercise physiology—publishable regardless of sign. A null result across the 0.2–3 mT scan would bound the in-vivo effect below the few-percent model prediction, which is itself a useful, honest constraint. Either way the value is data, obtained with a verified field and a blinded control—the opposite of the peak-reading, uncalibrated “free-energy” tradition this project deliberately left behind.

References

- [1] O. Kirichenko, “Chirality-induced spin selectivity as a spin gate on mitochondrial superoxide production: a radical-pair model,” companion note (this folder).
- [2] R. J. Usselman *et al.*, “The quantum biology of reactive oxygen species partitioning impacts cellular bioenergetics,” *Sci. Rep.* **6** (2016) 38543.
- [3] M. Ristow *et al.*, “Antioxidants prevent health-promoting effects of physical exercise in humans,” *PNAS* **106** (2009) 8665.

- [4] M. C. Gomez-Cabrera *et al.*, “Oral administration of vitamin C decreases muscle mitochondrial biogenesis and hampers training-induced adaptations in endurance performance,” *Am. J. Clin. Nutr.* **87** (2008) 142.